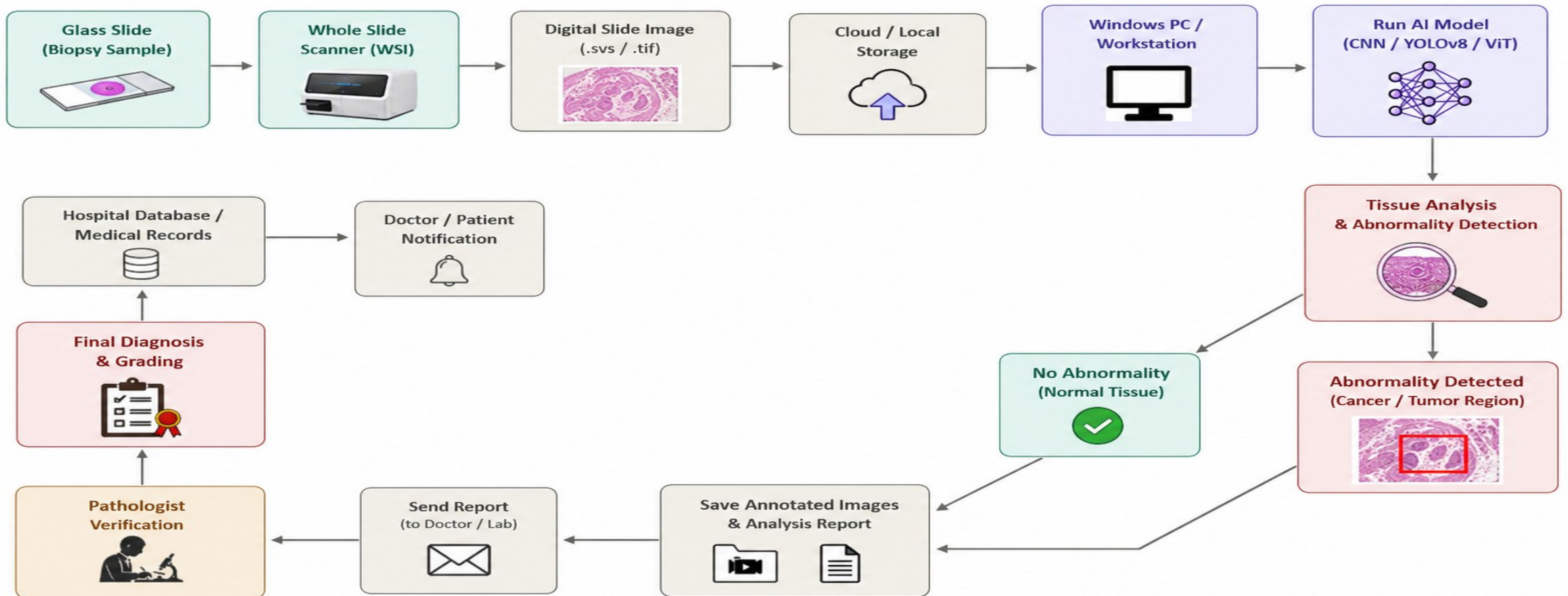


AT2 (CO1, 20%): Conceptual Architecture Mapping of Computer Vision Pipelines across Real-World Application Domains

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DIGITAL PATHOLOGY SLIDE ANALYSIS - SYSTEM ARCHITECTURE

Digital Pathology Slide Analysis – system architecture



DIGITAL PATHOLOGY SLIDE ANALYSIS

Digital Pathology Slide Analysis is an image processing and artificial intelligence (AI)-based system that analyzes high-resolution digital images of tissue slides to assist pathologists in detecting diseases accurately and efficiently. The main objective is to identify abnormalities such as cancer cells, tumors, inflammation, and other tissue disorders at an early stage, enabling faster and more reliable diagnosis. The system begins by scanning glass biopsy slides using a Whole Slide Imaging (WSI) scanner to create high-resolution digital pathology images. These images are then preprocessed through noise removal, color normalization, image enhancement, and tissue segmentation to improve analysis quality. Advanced image processing techniques such as feature extraction, image segmentation, and region-of-interest detection are applied to highlight important tissue structures. Deep learning models, such as Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), or YOLO-based architectures, are used to classify tissue as normal or abnormal and detect suspicious regions. The detected abnormalities are highlighted on the digital slide, and annotated images along with diagnostic reports are generated for pathologist review and verification. After confirmation, the final pathology report is stored in the hospital database and shared with healthcare professionals for treatment planning. This approach provides faster, more accurate, and cost-effective pathological diagnosis compared to traditional manual microscopy. It enhances diagnostic accuracy, reduces workload for pathologists, supports early disease detection, and ultimately improves patient care and clinical decision-making.